

Surname and Initials:	Session Day:	Group no
Student number:		



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

DEPARTMENT OF PHYSICS

FSK 116

Practical Report

PAR – Projectile Motion

FSK 116 Online Practical Activity: Projectile motion

Total: 40 marks.

Instructions:

The data provided in the practical, is read data captured by means of the *Tracker* programme from a video I made at home using my cell phone. The idea is that you use this data, but if you are more adventurous, download the programme from <https://physlets.org/tracker/> (free, open source) and make your own.

Follow the instructions below and upload your answers as a PDF with the answers clearly numbered.

Apparatus

Optional: Tracker (downloadable from <https://physlets.org/tracker/> or, using free data via <https://connect.up.ac.za/https/drive.google.com/file/d/1z2P3W99UcyrTD3WXwvy8P8bXyCSthJvs/view?usp=sharing>) installed on a computer, a cell phone, dark coloured ball, a well-lit white wall, four markers with Prestik, a tape measure. Graph paper or spreadsheet such as Excel or Google Sheets (Google sheets is available online using free data as <https://connect.up.ac.za/https/docs.google.com/spreadsheets/>) If this link does not work, try copying it and pasting it in directly on your browser or log into Gmail from the connect.up.ac.za portal, and then go to “Google Apps” and select “Sheets”)

The video is available online as:

Direct: <https://drive.google.com/file/d/1Q3wNT42ZVv5jmikNe0ErmmXhermWvh2-/view?usp=sharing>

Through connect: <https://connect.up.ac.za/https/drive.google.com/file/d/1Q3wNT42ZVv5jmikNe0ErmmXhermWvh2-/view?usp=sharing>

The data as a spreadsheet is available as:

Direct: <https://drive.google.com/file/d/1u1NTBhv0JGdyyJC1lA832eZEIPJfINb-/view?usp=sharing>

Through connect: <https://connect.up.ac.za/drive.google.com/file/d/1u1NTBhv0JGdyyJC1lA832eZEIPJfINb-/view?usp=sharing>

PART A: Capturing the data (optional, no marks, but a lot of fun 😊)

[A1] Optional, but fun: I did this from scratch with no experience, and it took two hours and was a lot of fun. Writing the instructions up took longer 😊. Note: Tracker tries its best to be user friendly – read the hints it gives on the screen.

- Download and install the Tracker software.
- Pick an evenly and well-lit, white wall. (I had a plant and some shadows from the washing line in the way. It is not ideal, but one can work around it.)
- Find a place where you can launch the ball from so that the ball travels in front of the wall for a considerable part of its trajectory. Ideally, it has to reach its maximum height and drop down to the level that you released it from.
- For calibration make two (removable!) marks on the wall a fixed horizontal distance apart. I stuck plastic wheels to the wall using Prestik. Do the same for the vertical calibration. (I picked 1 m for convenience.)
- Place your cell phone at a point where the whole motion can be recorded, preferably facing the wall perpendicularly.
- Record a video of the ball flying. (If possible, set the shutter speed to as short as possible – on some phones it means setting the ISO number high, using a high frame rate (30 fps) and a high resolution.)
- Open Tracker. Click on “Video”->“Import” and import your video.
- You probably recorded far too much, so use the small black triangles beneath the timeline to select the region of interest from the frame showing you releasing the ball to the frame just before the ball hits the ground. (You can go forward or backwards in steps of one frame at a time).
- If your video has interference like plants on it, right click and select “Filters”->“New”->“Negative” and “Filters”->“New”->“Baseline” (capture a frame without the ball) to subtract everything leaving you with just the ball (and in my case, its shadow) in white moving across a black screen.
- Click on “Track”->“New”->“Point Mass”
- Go to frame 1, and Ctrl-Shift-Click on the ball to mark it. Zoom in to define the template of the ball clearly (circle) and the search area (rectangle) to include the box.
- On the autotracker, click “Search This” to mark and select the ball.
- On the autotracker, click “Search Next”. If all went well, Tracker will have found the ball on the next frame and marked it (there will be a report in green text). If not, there will be a report in red text. Help autotracker to find the ball by moving the search area, or redefining the template (Ctrl-Shift-Click). If all works well, Tracker should track all the frames with maybe one or two exception.
- Repeat clicking “Search Next” until the ball is found on all frames. (The success of this depends on the quality of your video. I had plants in the way so I selected the filters “invert” and “baseline” to subtract everything that was not moving. Play around with the search area and evolution rate to make it happen. Note: You can zoom in to make it easier.)
- Select “Track”->“New”->“Coordinate System”. Move the coordinate system so that its origin is at the point where the ball is released, and the axes are horizontal and vertical.

- p) Select “Track”->”New”->”Calibration Tools”->”Calibration Stick” and Shift-Click on two of your calibration marks. Enter the distance between the two points to calibrate the video.
- q) Repeat above for your other two calibration points.
- r) You should now end up with a table of $t(s)$, $x(m)$ and $y(m)$ for your ball as well as graphs of $y(t)$ and $x(t)$ as shown below.
- s) You can also get tracker to display velocity and acceleration vectors which you can scale to look pretty.

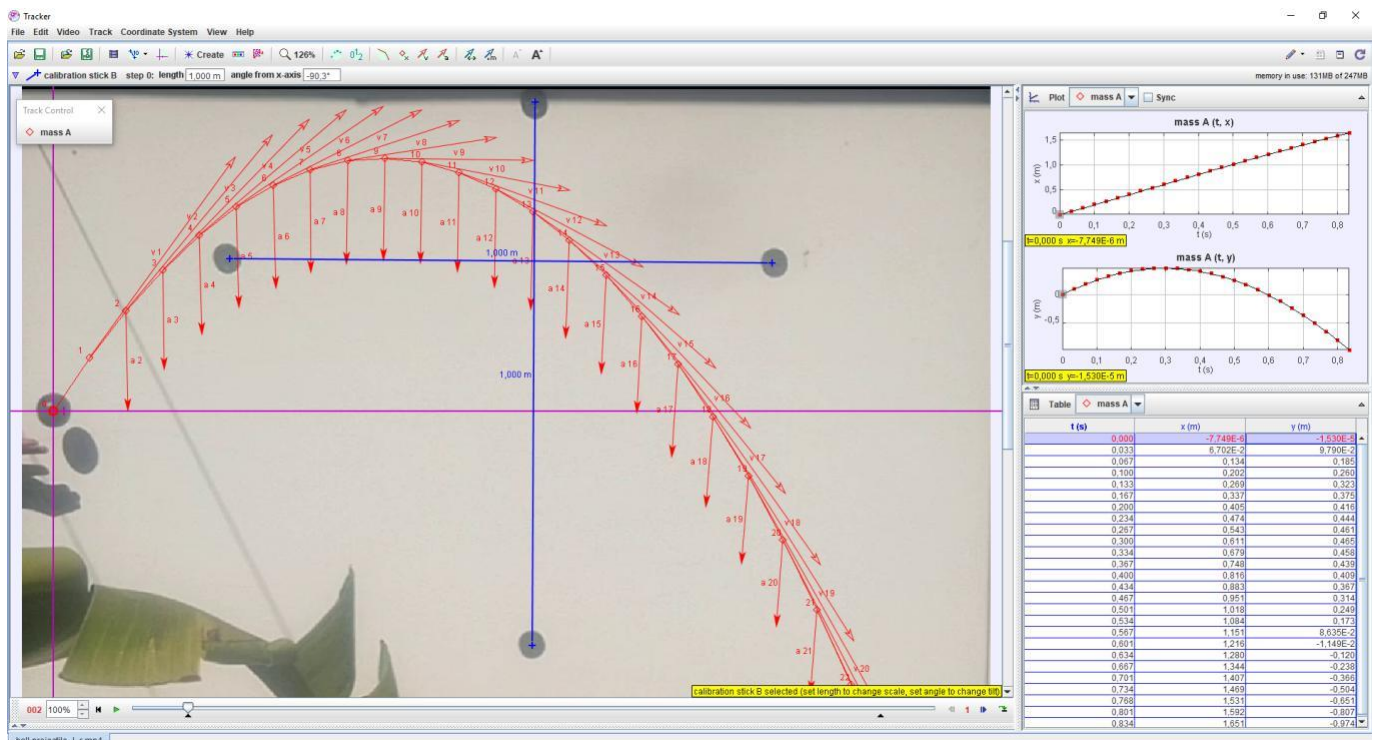


Figure 1: Screen shot of Tracker after capturing the whole motion. The blue lines are the “calibration sticks” that calibrate the distance between the markers. The numbers next to the points indicate the frame number, and v and a indicate the calculated velocity and acceleration vectors. On the right are graphs of $x(t)$ and $y(t)$ as well as a table with the (calibrated) raw data.

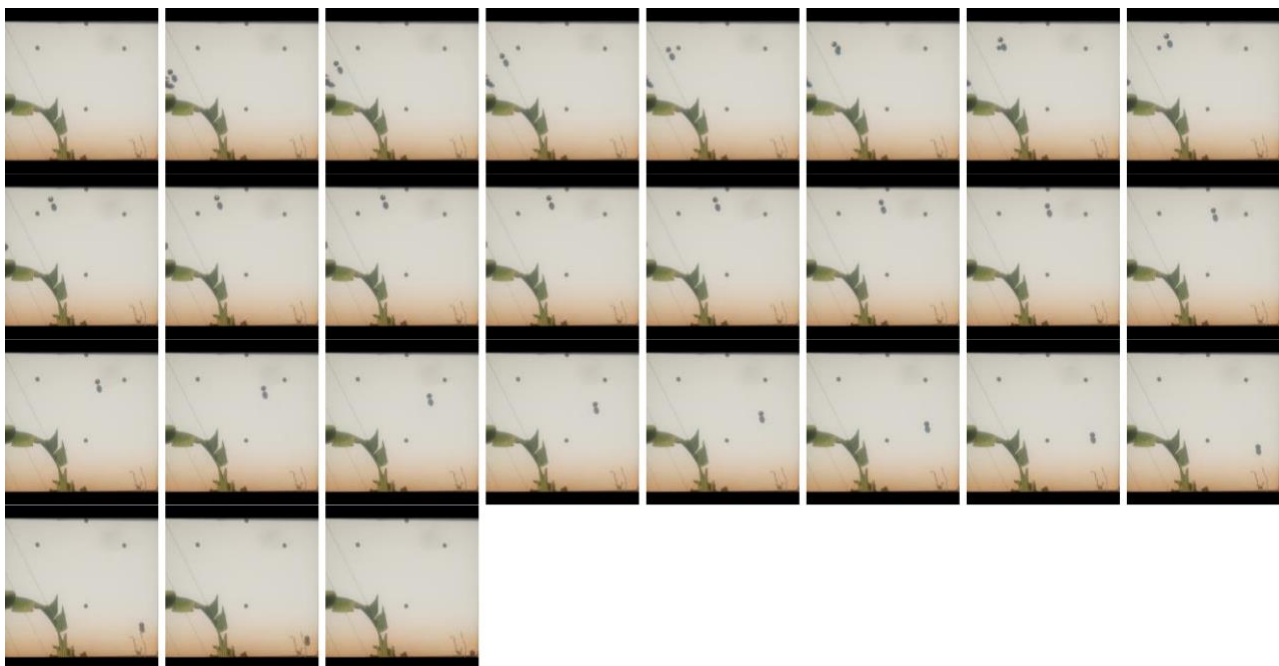


Figure 2: The 28 frames from which the data was collected at a frame rate of 30 frames per second. Note the four dots in the shape of a cross (length of arms 1.00 m) that were used to calibrate the length scale. The ball travels from the left to the right, and is most easily identified by the shadow it casts on the wall. In principle, one could use a ruler to find the x - and y -coordinates of the ball and apply the scaling to get to the data below.

PART B: Data analysis

Below is the captured data.

Table 1: Position and time data as captured by Tracker for a ball that was thrown.

t (s)	x (m)	y (m)
0.000	0.000	0.000
0.033	0.067	0.098
0.067	0.134	0.185
0.100	0.202	0.260
0.133	0.269	0.323
0.167	0.337	0.375
0.200	0.405	0.416
0.234	0.474	0.444
0.267	0.543	0.461
0.300	0.611	0.465
0.334	0.679	0.458
0.367	0.748	0.439
0.400	0.816	0.409
0.434	0.883	0.367
0.467	0.951	0.314
0.500	1.018	0.249
0.534	1.084	0.173
0.567	1.151	0.086
0.600	1.216	-0.011
0.634	1.280	-0.120
0.667	1.344	-0.238
0.701	1.407	-0.366
0.734	1.469	-0.504
0.768	1.531	-0.651
0.801	1.592	-0.807
0.834	1.651	-0.974

[B1] Plot graphs of $x(t)$, $y(t)$ and $y(x)$.

Computer: If you use a computer, please use all points.

By hand: If you plot them by hand, just use the points from 0 to 0.6 s, in 0.1 s intervals (as indicated in bold). Graph paper is included at the end of this document.

Very important: Ensure that your graphs comply with the requirements of a scientific graph as we discussed in Experiment 1 and 2 (GRA)

GRAPHS

Hint: Fit a quadratic curve to the graph of $y(t)$ and compare its coefficients with the expected experimental values. (Google graphs can do it. Go to “Chart Style”->”Series” and select “Trend Line”. There you can select a quadratic graph and there is an option to print the equation.

(If using a computer, plot all points, if plotting by hand, use the same points as previously.)

[B4] Using the results obtained in the previous section, calculate $a_{y,avg}(t) = v_{y,avg} / t$, present your results in a table. Note that the best time to associate with the average acceleration is the midpoint of the intervals used in B3 above.

[B4] Plot graphs of $v_x(t)$ and $v_y(t)$ and $a_y(t)$. Also calculate the average acceleration of the ball during its flight. (*Remember rules for drawing graphs!*)

[illegible]

TABLE 2

Title: _____

			<i>Midpoint of time interval</i>	<i>Average velocity</i>	<i>Average velocity</i>	<i>Midpoint of time interval</i>	<i>Average acceleration</i>
<i>t</i> (s)	<i>x</i> (m)	<i>y</i> (m)	_____	_____	_____	_____	_____
0.000	0.000	0.000					
0.033	0.067	0.098					
0.067	0.134	0.185					
0.100	0.202	0.260					
0.133	0.269	0.323					
0.167	0.337	0.375					
0.200	0.405	0.416					
0.234	0.474	0.444					
0.267	0.543	0.461					
0.300	0.611	0.465					
0.334	0.679	0.458					
0.367	0.748	0.439					
0.400	0.816	0.409					
0.434	0.883	0.367					
0.467	0.951	0.314					
0.500	1.018	0.249					
0.534	1.084	0.173					
0.567	1.151	0.086					
0.600	1.216	-0.011					
0.634	1.280	-0.120					
0.667	1.344	-0.238					
0.701	1.407	-0.366					
0.734	1.469	-0.504					
0.768	1.531	-0.651					
0.801	1.592	-0.807					
0.834	1.651	-0.974					

Average acceleration = _____

GRAPHS

PART C: Vectors:

Position, displacement and average velocity.

[C1] In unit vector notation, write down the position vector of the ball at $t = 0.3, 0.4$ and 0.5 s. [Call them $\mathbf{r}(0.3)$, $\mathbf{r}(0.4)$ and $\mathbf{r}(0.5)$ respectively]. Use values from Table 1 above.

[C2] Now calculate $\mathbf{r}_{t=0.3 \text{ to } 0.4 \text{ s}}$ and $\mathbf{r}_{t=0.4 \text{ to } 0.5 \text{ s}}$. Again, write each of these in unit vector notation, and do not forget vector lines or units!

[C3] Calculate the approximate instantaneous velocity of the ball at 0.35 and 0.45 s using $\mathbf{v}(0.35) \approx \mathbf{r}_{t=0.3 \text{ to } 0.4 \text{ s}} / t$ and $\mathbf{v}(0.45) \approx \mathbf{r}_{t=0.4 \text{ to } 0.5 \text{ s}} / t$.

[illegible]

[illegible]

[C7] Comment on your answer in [C6] and how well it corresponds to theory.

PART D: Discussion:

Are the results obtained in this experiment 100% accurate? List the aspects that you think could contribute to answers deviating from the ideal.

GRAPH PAPER

